Toolbox Card

Eight tools for starting a problem

AGES 10-13



1 Spot the pattern

Find the rule, test it on a new case, then use it to jump far.



2 List every way

Count every case once with a systematic order; rule out repeats.



3 Draw it

Sketch and label a model; let the picture show what stays the same.



4 Make a table

Tabulate cases, find the change in each column, extend or generalise.



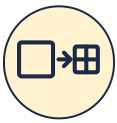
5 Work backwards

Reverse the operations from the result, then confirm by running forwards.



6 Guess and check

Guess, check, adjust using how far off you were; stop when every condition holds.



7 Try something simpler

Solve 1, 2, 3 cases first, find how the answer grows, then handle the real size.



8 Act it out

Model or simulate the situation (objects, fold, cut, dice) and record the outcomes.

Look back

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10-13

Pick one problem from today. There are no right answers. You can pass. Nobody reads it out.

1. **Did I answer the question that was asked (with units or a reason)?**
2. **Is my answer reasonable? (Estimate or compare.)**
3. **How can I check it another way? (Go backwards, estimate or try a smaller case.)**

I checked by ... Use a pencil.

When I am stuck

- **Wait.**
- **Try a tool.**
- **Try a smaller case.**
- **Ask.**

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